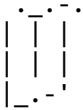


```html

# 🌟 GSAP: GreenSock Animation Platform Study Notes 🌟

## Everything About GSAP: JavaScript Animation Library



### 1. Introduction to the Series & GSAP 🎬

- This series, "**Everything About GSAP**," aims to cover comprehensive details of the GreenSock Animation Platform.
- **GSAP (GreenSock Animation Platform)** is a powerful JavaScript library for creating animations.
- **Key Topics Covered:**
  - GSAP Basics
  - ScrollTrigger 📄
  - SVG Animation 🎨
  - Five different practical projects 🚀
- **Video Sequence:** Basics → ScrollTrigger → SVG Animation → Project 1 → Project 2 → Project 3 → Project 4 → Project 5.
- **Schedule:** GSAP videos are expected around 11-12 PM daily.
- **Goal:** To provide a detailed, engaging, and user-friendly learning experience.

### 2. What is Animation? 🤔

- **Definition:** On web pages and websites, **any movement, transition, or effect** can be termed an animation.
- **Purpose/Benefits:**
  - Makes the user more **engaged**.
  - **Enhances user experience**.
  - Encourages users to spend **more time** on your website.
- **Examples:**
  - A simple **hover effect** when the mouse moves over a button.
  - A car moving across the screen in a web animation.
- Animations are visual effects that make websites more **attractive**.

### 3. What Exactly is GSAP? 🧠

- **Full Form: GreenSock Animation Platform.**
- **Type: A JavaScript animation library.**
- **Functionality:** Helps in creating complex animations with significantly **less code** and **more features**.
- **Prerequisite:** While GSAP is a JavaScript library, this series is designed so you can follow along **even without prior JavaScript knowledge**. The syntax is slightly different but taught clearly.

## 4. Understanding Libraries

- **Definition:** Libraries basically provide a **specific type of code or feature**.
- **Analogy:** Similar to having a pipeline from a rooftop water tank – you don't bring the whole tank down; you just access its functionality via a connection.
- **Examples of Libraries and their Features:**
  - **Locomotive.js:** Used for **smooth scrolling** on websites.
  - **GSAP:** Used for **moving animations** (moving elements, transitions, effects related to movement).
  - **React.js:** Used for building entire **frontends and UIs**.
  - **ScrollTrigger (GSAP Plugin):** Handles **effects and animations based on scrolling**.
  - **Swiper.js:** Creates **sliding animations**.
- Each library offers a particular feature that adds a unique characteristic to your website.

## 5. Getting Started with GSAP

### 5.1. GSAP Website

- Visit the official GSAP website: [greensock.com](https://greensock.com).
- Current tagline: "Animate Anything. A widely robust JavaScript animation library built for professionals."

### 5.2. Project Setup

- Create a folder on your desktop, preferably named "**GSAP**".
- Open this folder in **VS Code**.
- Create the following files:
  - ``index.html``
  - ``style.css``
  - ``script.js``
- **Boilerplate HTML:** Use the standard HTML boilerplate in ``index.html``.
- **Link Files:**
  - Link ``style.css`` in the ```` section of ``index.html``.
  - Link ``script.js`` at the end of the ```` section of ``index.html``.

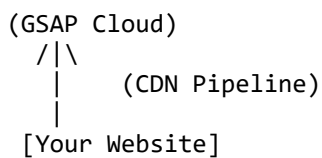
### 5.3. Creating a Test Element

- In ``index.html``, add a simple ``div`` with an ID:

- ```
<div id="box"></div>
```
- In `style.css`, style the box:

```
#box {
  height: 300px;
  width: 300px;
  background-color: crimson;
}
body {
  background-color: #333; /* For better visibility */
}
```
 - Open `index.html` with Live Server to see the box.

5.4. Linking GSAP via CDN (Content Delivery Network)



- **Purpose of CDN:** To establish a connection between your website and GSAP's original code (hosted on the cloud) without downloading the entire library locally.
- **Steps:**
 1. Search "GSAP CDN" on Google.
 2. Go to the first result (typically `cdnjs.com` or `greensock.com`'s CDN section).
 3. Find and **copy the** `